

Classical Planning Report

Table 1 (Problems 1 and 2, all 11 searches)

Actions, expansions, plan length, time in seconds.

Problem	Actions	Search	Heuristic	Expansions	Plan length	Time (s)
P1	20	BFS		43	6	0.0013
P1	20	DFS		21	20	0.00070
P1	20	UCS		60	6	0.0020
P1	20	Greedy	unmet_goals	7	6	0.00037
P1	20	Greedy	levelsum	6	6	0.027
P1	20	Greedy	maxlevel	6	6	0.019
P1	20	Greedy	setlevel	6	6	0.081
P1	20	A*	unmet_goals	50	6	0.0019
P1	20	A*	levelsum	28	6	0.067
P1	20	A*	maxlevel	43	6	0.065
P1	20	A*	setlevel	33	6	0.182
P2	72	BFS		3,343	9	0.45
P2	72	DFS		624	619	0.57
P2	72	UCS		5,154	9	0.67
P2	72	Greedy	unmet_goals	17	9	0.0035
P2	72	Greedy	levelsum	9	9	0.59
P2	72	Greedy	maxlevel	27	9	0.88
P2	72	Greedy	setlevel	9	9	1.95
P2	72	A*	unmet_goals	2,467	9	0.50
P2	72	A*	levelsum	357	9	14.8
P2	72	A*	maxlevel	2,887	9	84.4
P2	72	A*	setlevel	1,037	9	175.2

S indices: 1 BFS, 2 DFS, 3 UCS, 4–7 greedy (unmet_goals / levelsum / maxlevel / setlevel), 8–11 A* (unmet_goals / levelsum / maxlevel / setlevel)

My picks for P3 and P4:

Role	Pick	Why
Uninformed	S1 BFS	Fastest optimal uninformed search on P1/P2 (DFS is faster but not optimal)
Greedy	S4 unmet_goals	Greedy is by far the fastest on P2.
Greedy	S5 levelsum	Fewest greedy expansions but still much faster than setlevel.
A*	S8 unmet_goals	Fastest A* on P1 and P2, and still found the short plans.
A*	S9 levelsum	Fewest A* expansions. A lot faster than maxlevel and setlevel on P2

Table 2 (Problems 3 and 4)

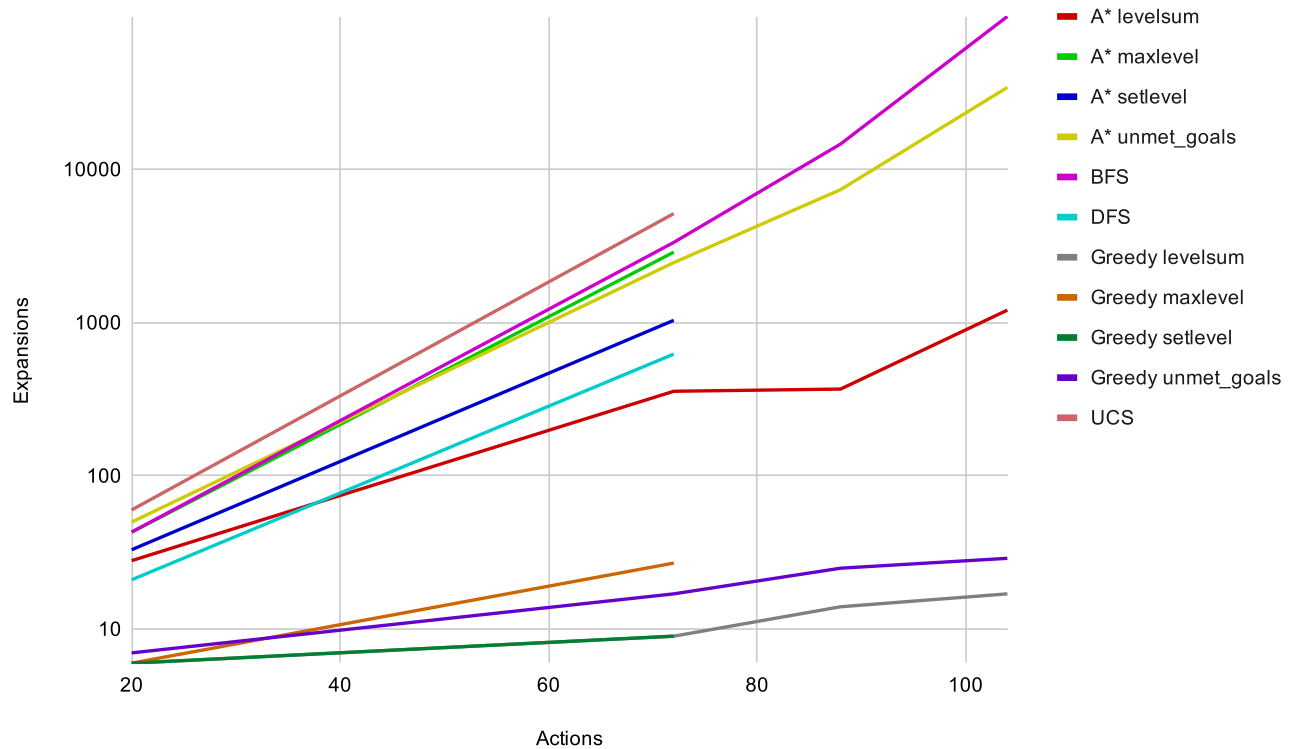
P3 has **88** actions; P4 has **104**. Runs: S1 BFS, S4 greedy + unmet_goals, S5 greedy + levelsum, S8 A* + unmet_goals, S9 A* + levelsum.

Problem	Actions	Search	Heuristic	Expansions	Plan length	Time (s)
P3	88	BFS		14,663	12	2.21
P3	88	Greedy	unmet_goals	25	15	0.0088
P3	88	Greedy	levelsum	14	14	1.32
P3	88	A*	unmet_goals	7,388	12	1.64
P3	88	A*	levelsum	369	12	27.9
P4	104	BFS		99,736	14	20.3
P4	104	Greedy	unmet_goals	29	18	0.014
P4	104	Greedy	levelsum	17	17	2.34
P4	104	A*	unmet_goals	34,330	14	11.1
P4	104	A*	levelsum	1,208	15	156.5

BFS and A* + unmet_goals both found a length of 12 (P3) and 14 (P4). Both greedys were longer. A* + levelsum matched 12 on P3 but returned 15 on P4.

Chart 1 (Expansions):

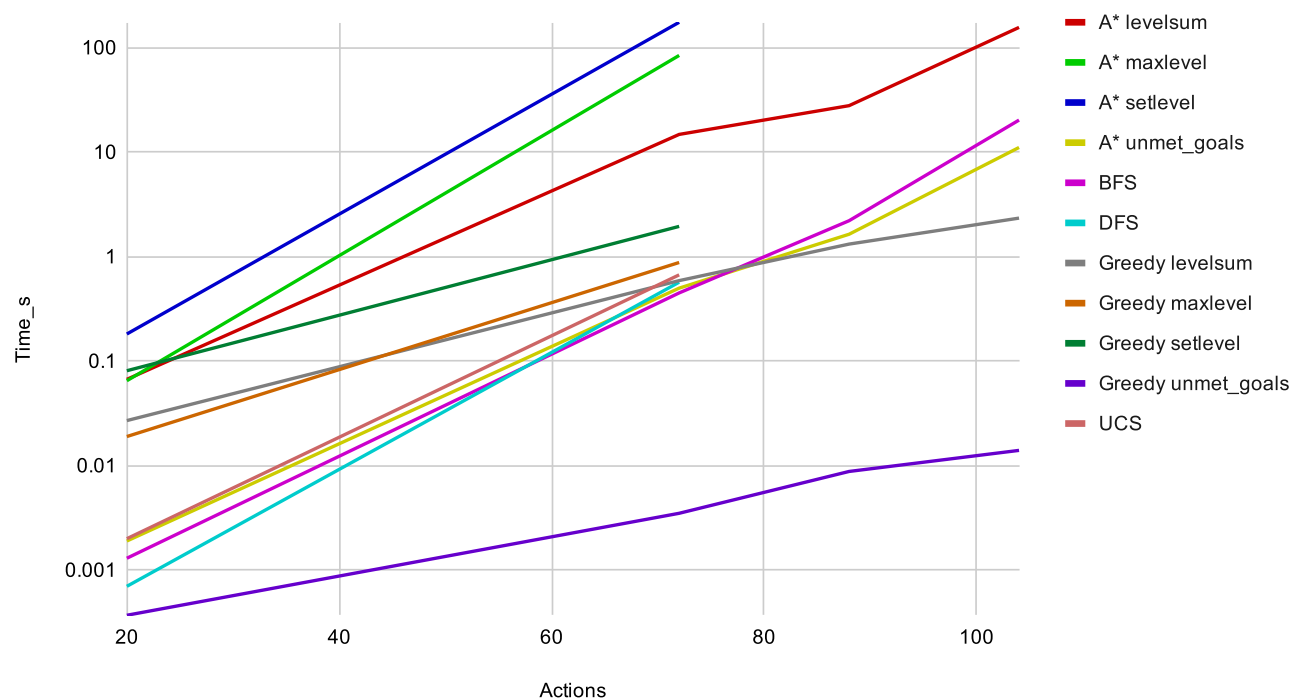
Expansions by Actions



Expansions grow almost exponentially as actions go from 20 to 104, except greedy, which stays almost flat. BFS and A* + unmet_goals grow exponentially (BFS is about 100k on P4). A* + levelsum grows much less. Greedy levelsum and setlevel both have 6 expansions on P1 and 9 on P2, so those two lines overlap.

Chart 2 (Time in s):

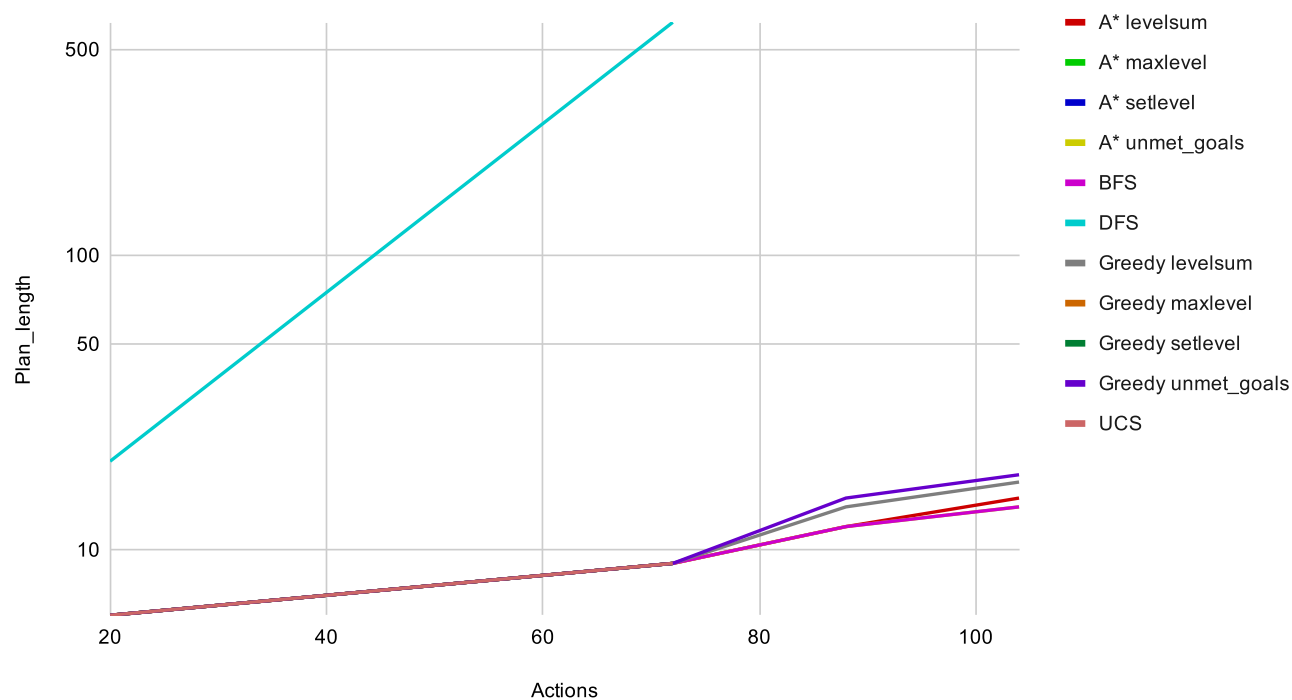
Time_s by Actions



Greedy + unmet_goals stays around 0.01 seconds even on P4. A* + levelsum grows exponentially and hits 156 seconds on P4; BFS ~20 seconds and A* + unmet_goals ~11 seconds.

Chart 3 (Plan Length):

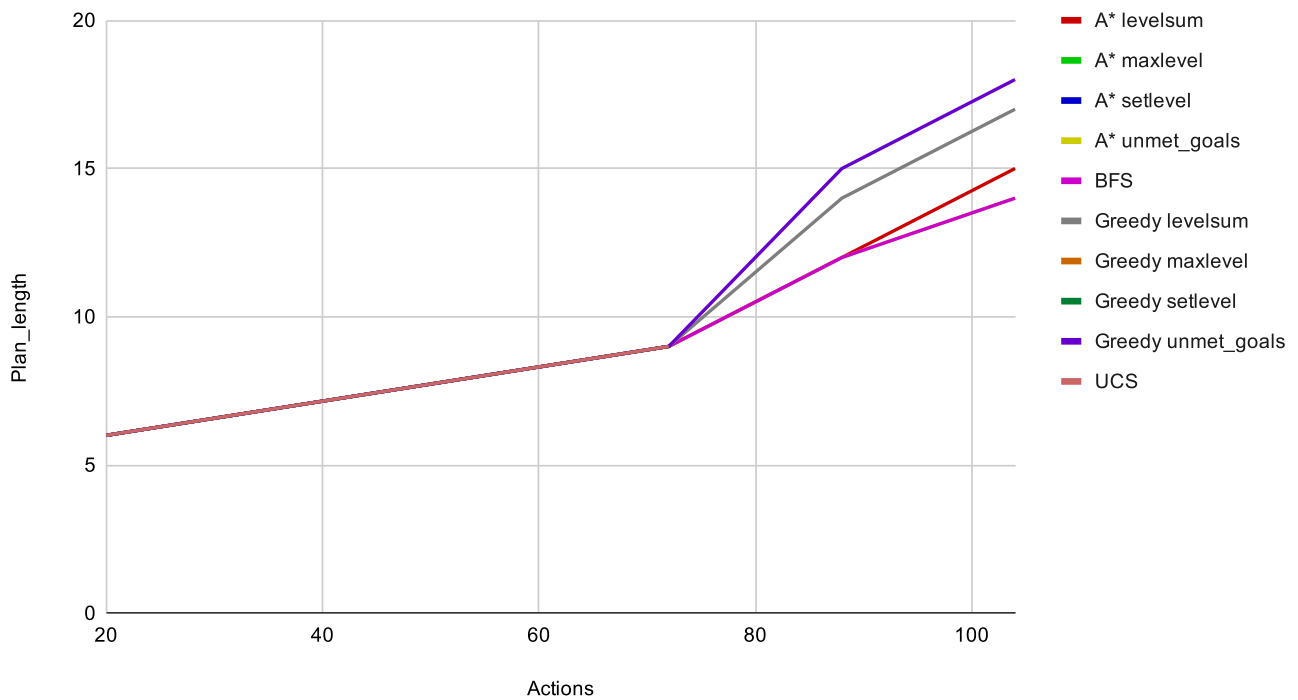
Plan_length by Actions



DFS plan length blows up (20 on P1, 619 on P2). On P1/P2 most other searches are length 6 then 9, so those lines overlap (greedy levelsum and setlevel included).

Chart 4 (Plan Length DFS omitted for clarity):

Plan_length by Actions



With DFS omitted the other algorithms stay close together on P1 and P2 (length 6 and 9). On P3/P4: BFS and A* + unmet_goals are the shortest lengths (12 and 14). Greedy has a longer plan length, and A* + levelsum did not find the shortest plan on P4 (15 instead of 14).

Questions

Which algorithm or algorithms would be most appropriate for planning in a very restricted domain (i.e., one that has only a few actions) and needs to operate in real time?

I would pick Greedy + unmet_goals. On P1 with 20 actions it was the fastest with 0.00037 seconds and still returned a length of 6. This is almost realtime for a restricted domain.

Which algorithm or algorithms would be most appropriate for planning in very large domains (e.g., planning delivery routes for all UPS drivers in the U.S. on a given day)

For planning the UPS delivery routes I would use Greedy + unmet_goals. It performed well on P4 with only 29 expansions and 0.014 seconds. While it did not find the shortest plan it is the algorithm best suited for our largest problem P4.

Which algorithm or algorithms would be most appropriate for planning problems where it is important to find only optimal plans?

For finding the optimal plan I would use BFS (found optimal plans 6, 9, 12, 14) and UCS (found optimal plans 6, 9). Both do not use heuristics. For the algorithms with heuristics I would use A* + maxlevel and A* + setlevel. The maxlevel and setlevel heuristics are admissible so A* is guaranteed to find the optimal plan.